Applications, URL-Categorisations & Reputations based Traffic Generation using Python Scripts

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Tools Used	Python3
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- Purpose
- Traffic Generation:
    - App-ID Filtering:
    - URL Filtering:
- Configuration Template:

Purpose

Lets assume we want to demonstrate the firewall restrictions based on pre-defined categories, reputations, Applications, Application-filters/Groups and so on. Finding applications/urls matching the appropriate categories like malware sites etc would be a tedious process. To reduce these efforts, this script enables us to simulate traffic matching different url-categories, reputations, application-groups etc matching the provided service templates config during the below scenarios:

1. Customer Demos and PoCs
2. SD-Security App/URLF testing
3. App-ID & URL Filtering for VSA/SASE use-cases
4. Quick Functional testing on new releases

The script is also designed in such a way that it enables users to simulate traffic based on the following options:

1. Applications

2. Application-Groups

3. Application-Filters - High-Risk, Non-Business & Web-Browsing

4. URL-Categories - Shopping, Social-Network, Job-search, Gambling, Adult Content, Malware sites, Weapons, dating etc.

5. URL-Reputations - Trustworthy, Moderate_Risk, Suspicious, High-Risk

6. Manual provisions for both Applications & URLs

Currently, the scripts are integrated with pre-defined urls that would simulate the required categories as required like malware, weapons sites etc.

There will also be a Service Template which can be cloned/exported to any environments for re-usability.

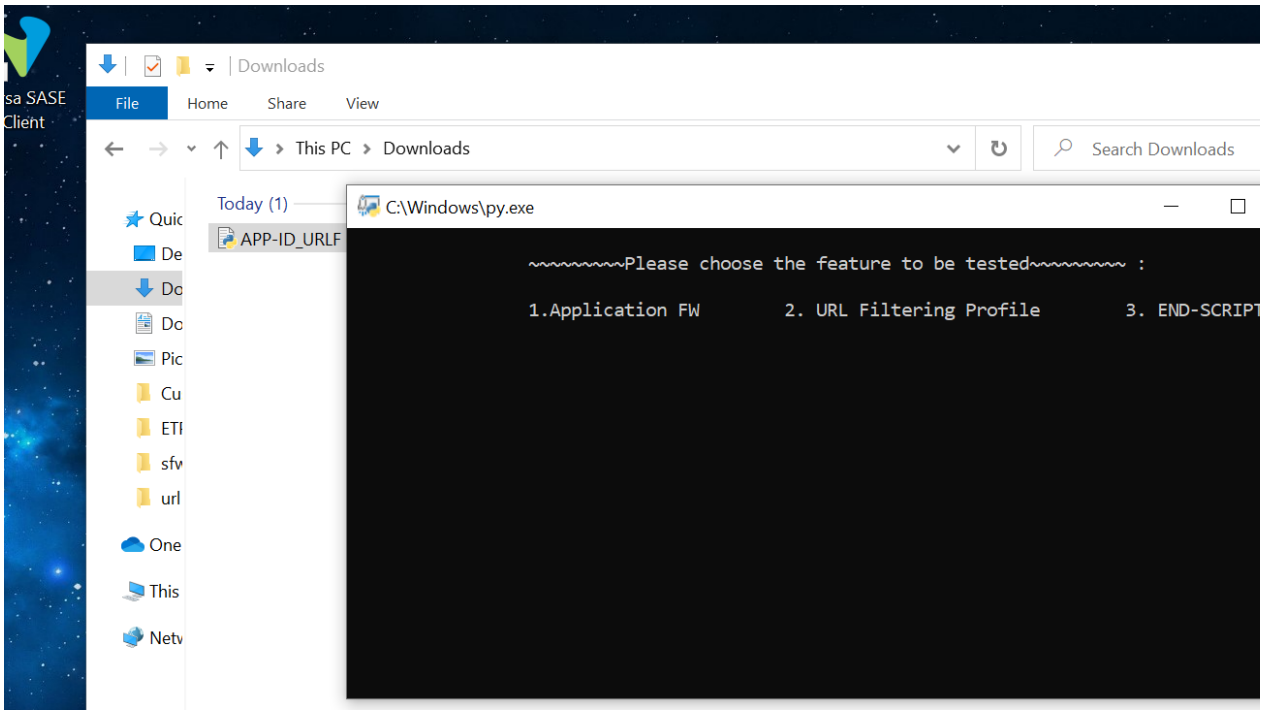
Traffic Generation:

Prerequisites:

- Following tools needs to be installed on Windows 10:

1. Python3
2. Web Browser (Google-Chrome)
3. VSA Client (for SASE use-cases)

- The Scripts can be run all popular OS like Ubuntu/Linux (or) Windows as long as Chrome and Python3 is installed.
- Traffic-generation script - APP-ID_URLF_Chrome.py
- Download the attached python file to a PC in your test environment from which you need to generate web-traffic
- Once copied, run the file using python3.
- This will prompt you to choose one of the traffic generation profiles - Application or URL Filtering, as shown below:

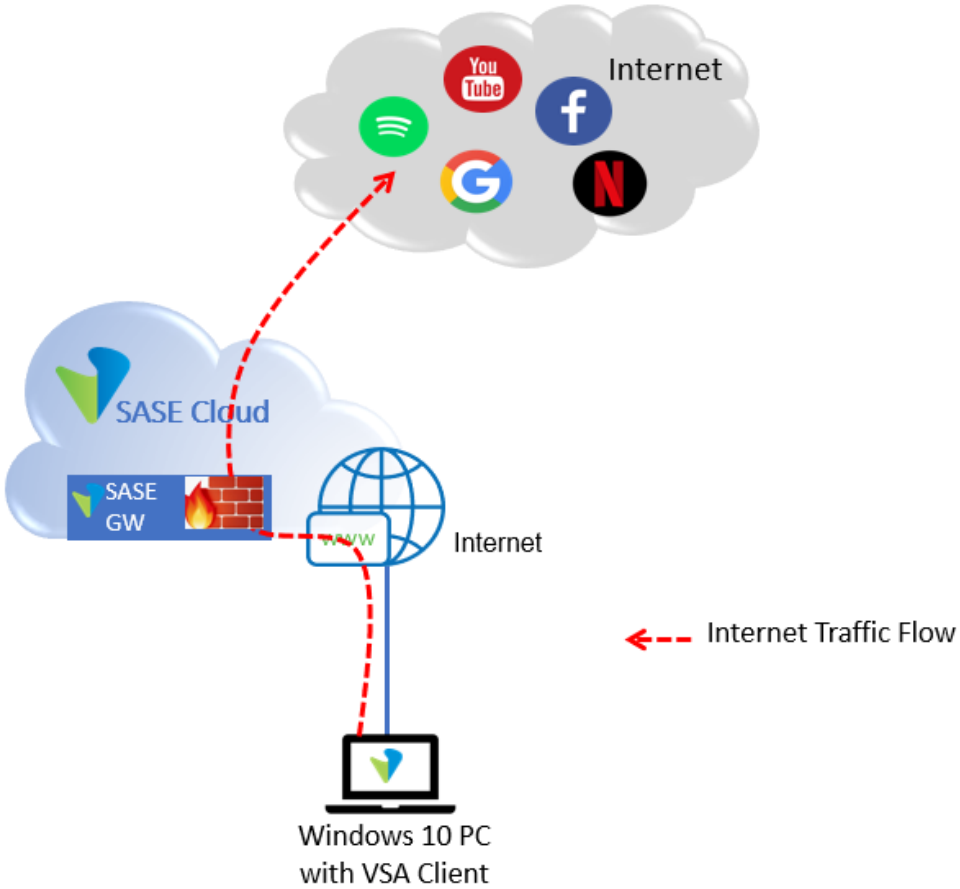


Example scenario used for this wiki section:

Here, for this script demonstration, we use a Windows PC installed with VSA client (7.5.6) & connected to a SASE gateway. The gateway is enabled with NGFW & has App-ID & URLF policies configured.

(Please refer the service templates attached later in this section to view the App-ID/URL-Filtering rules)

The purpose is to use the script to easily generate the desired web-application traffic from any Windows machine (LAN/VSA client) to verify the Next-Gen Firewall rule actions configured in the VOS device.



App-ID Filtering:

- For App-ID FW, run the python script and select "1"

```
~~~~~~~~~~Please choose the feature to be tested~~~~~~~~~~ :

1.Application FW        2. URL Filtering Profile        3. END-SCRIPT
```

- Selecting "1.Application FW" will take you to the below screen where you can either choose,
    1. Predefined list - Application/Application-Group/Application-filter traffic, (or)
    2. Manual input

```
~~~~~~~~~~Please choose the feature to be tested~~~~~~~~~~ :

1.Application FW        2. URL Filtering Profile        3. END-SCRIPT
1

Choose from : 1. Predefined: Applicaions/Applicaiton-Groups/Application-Filter list     2. Enter Web Application URL manually    3.Exit
```

- Under predefined Application/App-group/App-Filter list, you can choose one of the three options -
    1. Applications
    2. Application Groups
    3. Application Filter

```
Choose from : 1. Predefined: Applicaions/Applicaiton-Groups/Application-Filter list    2. Enter Web Application URL manually    3.Ex
1
1. Applications                2. Application Groups    3. Applicaiton Filter          4. Exit
```

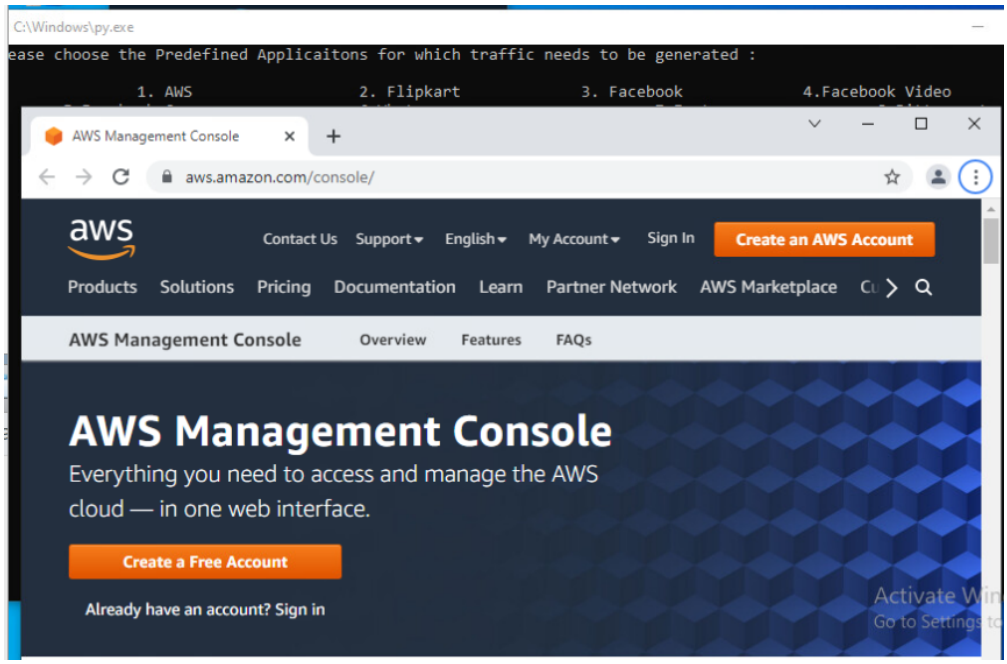
- Selecting "1.Applications" prompts you to choose a web-app from the list as shown below:

Note: O365 Applications requires login to Microsoft Account.

```
Please choose the Predefined Applicaitons for which traffic needs to be generated :

        1. AWS                  2. Flipkart             3. Facebook             4.Facebook Video
        5.Facebook Games        6.Whatsapp              7.Instagram             8.Bittorrent
        9.Kproxy                10.Google_Ads           11.Youtube              12.Netflix
        13.MS-Sharepoint        14.MS-Powerpoint        15.MS-Outlook           16.MS-Excel
        17.MS-Teams             18.Google-groups        19.Exit
```

- Choose any application from the list (1-18) as necessary. This opens the respective web-application in a new browser tab.

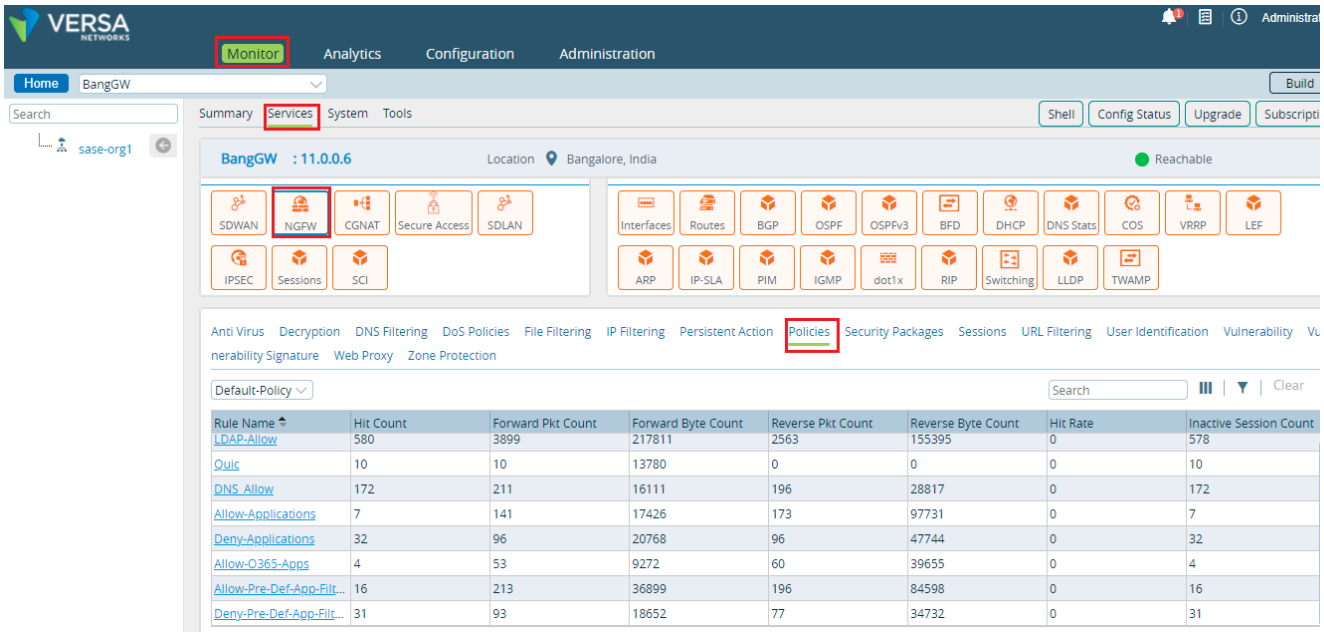
  For example, selecting "1.AWS" will open AWS console in a new Chrome tab as below



NOTE: If any additional requirements required, please leave a comment to check the feasibility to accommodate into the script.

As per this set-up, the web-app AWS-console is set to 'allow' and hence the webpage accessible.

- Firewall Rule hit-counts can be viewed from Director GUI as shown below, Under **Monitor -> Services -> NGFW -> Policies** of VOS gateway device :



- Application/Application-Groups/Application-Filter statistics can be viewed from the VOS Device using below commands.

"show orgs org-services <Org-Name> application-identification statistics application brief"

```
admin@GW-1-cli> show orgs org-services sase-org1 application-identification statistics application brief
                                                                    TOTAL     TOTAL     TOTAL      TOTAL
                                                                    PACKETS   PACKETS   BYTES      BYTES
NAME                 ID      TYPE          SESSIONS  TRANSACTIONS   FORWARD   REVERSE   FORWARD    REVERSE
--------------------  ------  ------------  --------  ------------  --------  --------  ----------  --------
01NET                1549    predefined    1         1             91        138       10283       118938
ADFORM               2706    predefined    12        12            744       1064      73243       748293
ADOBE                2199    predefined    15        15            317       362       83112       155086
ADVERTISING_COM      2201    predefined    1         1             21        22        4377        11182
AMAZON               808     predefined    49        49            5027      7506      1056462     7118997
AMAZON_ADSYSTEM      1911    predefined    6         6             191       224       58529       127739
AMAZON_AWS           809     predefined    16        16            1057      1623      121518      1815610
APPNEXUS             2218    predefined    1         1             18        20        4331        9959
BIDSWITCH            2746    predefined    1         1             20        28        4558        11986
BING                 59      predefined    18        18            384       398       92574       209607
BITLORD              1926    predefined    8         8             129       159       16653       94143
BITS                 846     predefined    4         4             161       212       16348       216332
BRIGHTCOVE           2235    predefined    2         2             48        68        6130        60444
CARTOONNETWORK       1296    predefined    7         7             97        103       13538       44813
CHROME_UPDATE        79      predefined    100       100           1102      977       113345      486050
CISCO                3359    predefined    1         1             67        101       9830        105399
CLOUDFLARE           1464    predefined    1         1             29        39        3266        26488
DAILYMOTION          90      predefined    15        15            725       1189      79972       1304756
DNS                  108     predefined    1212      1212          1535      1395      117046      187460
```

"show orgs org-services <Org-Name> application-identification statistics predefined-application-group"

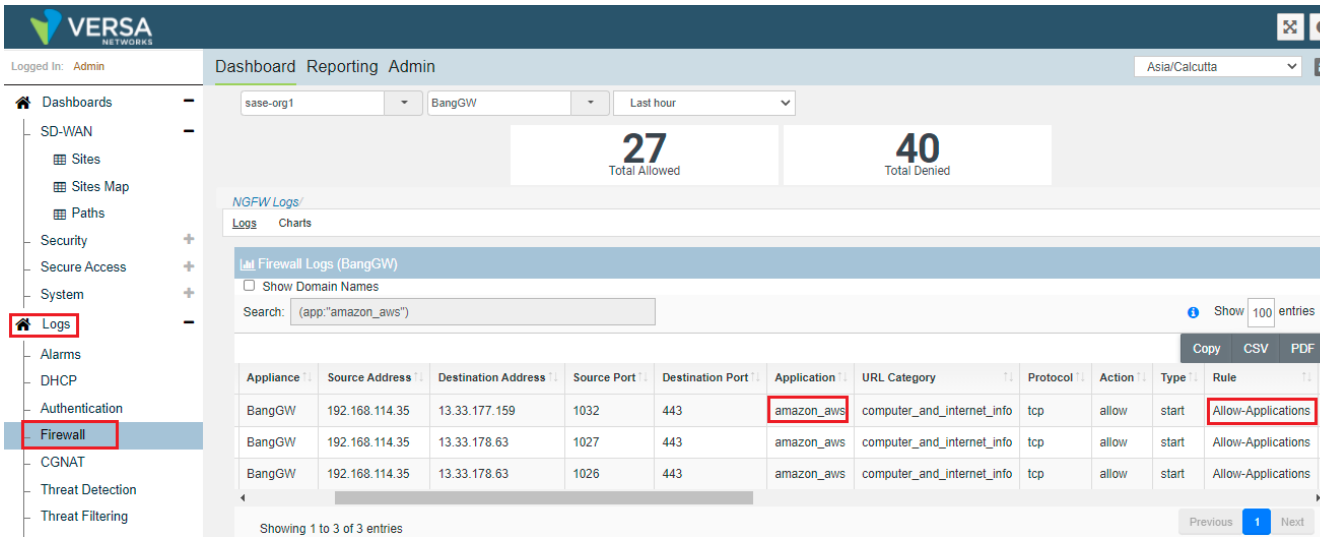
```
admin@GW-1-cli> show orgs org-services sase-org1 application-identification statistics predefined-application-group
                                                                                                TOTAL     TOTAL     TOTAL     TOTAL
                                      TOTAL     TOTAL     TOTAL     TOTAL                       PACKETS   PACKETS   BYTES     BYTES
NAME              SESSIONS  TRANSACTIONS  PACKETS   PACKETS   BYTES     BYTES     SESSIONS  TRANSACTIONS  FORWARD   REVERSE   FORWARD   REVERSE   TOTAL
                                          FORWARD   REVERSE   FORWARD   REVERSE   RATE      RATE          RATE      RATE      RATE      RATE      BYTES
-------------------------------------------------------------------------------------------------------------------------------------------------------------
ADP-Apps          0         0             0         0         0         0         -         -             -         -         -         -         0
Adobe-Apps        0         0             0         0         0         0         -         -             -         -         -         -         0
Amazon-Apps       71        71            6275      9353      1236509   9062346   -         -             -         -         -         -         10298855
Box-Apps          0         0             0         0         0         0         -         -             -         -         -         -         0
Citrix-Apps       0         0             0         0         0         0         -         -             -         -         -         -         0
Concur-Apps       0         0             0         0         0         0         -         -             -         -         -         -         0
Docusign-Apps     0         0             0         0         0         0         -         -             -         -         -         -         0
Dropbox-Apps      0         0             0         0         0         0         -         -             -         -         -         -         0
Google-Apps       8         8             631       667       60333     193009    -         -             -         -         -         -         253342
GotoMeeting-Apps  0         0             0         0         0         0         -         -             -         -         -         -         0
```

"show orgs org-services <Org-Name> application-identification statistics predefined-application-filter"

```
admin@GW-1-cli> show orgs org-services sase-org1 application-identification statistics predefined-application-filter
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
                                                                                                                                                                                                                                                                                                                                                                                                                                                
```

NAME	SESSIONS	TRANSACTIONS	TOTAL PACKETS FORWARD	TOTAL PACKETS REVERSE	TOTAL BYTES FORWARD	TOTAL BYTES REVERSE	SESSIONS RATE	TRANSACTIONS RATE	TOTAL PACKETS FORWARD RATE	TOTAL PACKETS REVERSE RATE	TOTAL BYTES FORWARD RATE	TOTAL BYTES REVERSE RATE	TOTAL BYTES
Advertising	60	60	3052	3516	337094	1426590	-	-	-	-	-	-	1763684
Audio-Video-Streaming	11	11	7933	13788	570530	18119822	-	-	-	-	-	-	18690352
Business-Traffic	1166	1166	15710	20804	1739156	22757078	-	-	-	-	-	-	24496234
Database	0	0	0	0	0	0	-	-	-	-	-	-	0
File-Transfer	1577	1577	22386	29149	2303901	27606353	-	-	-	-	-	-	29910254
Gaming	0	0	0	0	0	0	-	-	-	-	-	-	0
High-Risk-Applications	786	786	5239	4424	335643	1803051	-	-	-	-	-	-	2138694
IOT	0	0	0	0	0	0	-	-	-	-	-	-	0
Mobile-Traffic	175	175	3189	3967	444368	1549376	-	-	-	-	-	-	1993744
Non-Business-Traffic	465	465	17726	22807	2677692	16679397	-	-	-	-	-	-	19357089
P2P	8	8	129	159	16653	94143	-	-	-	-	-	-	110796
SaaS-Applications	216	216	9514	11606	1181337	7403782	-	-	-	-	-	-	8585119
Software-Updates	264	264	3678	4513	387867	2573570	-	-	-	-	-	-	2961437
VOIP	7	7	114	133	22167	76590	-	-	-	-	-	-	98757
Web-Browsing	925	925	48195	62897	8329440	53205818	-	-	-	-	-	-	61535258

- Firewall Logs can be viewed from the Analytics (provided LEF is configured)



- Similar procedure applies for Application-Groups & Application-Filter as well
- Under "2.Application-Groups", we have the below predefined options:

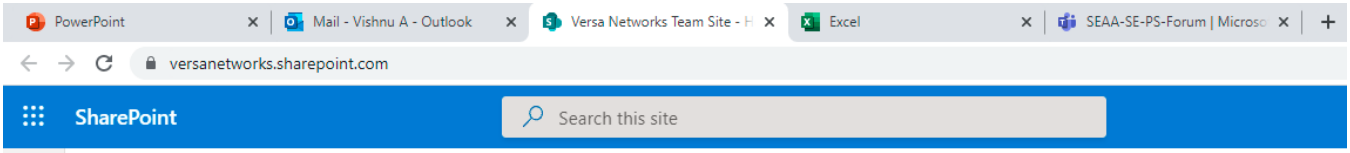
1. Amazon-Apps
2. Google-Apps
3. Social-Media-Apps
4. Office365-Apps (Requires Login to Microsoft account)

```
                1. Applications                2. Application Groups    3. Applicaiton Filter          4. Exit
2

Please choose the Predefined Applicaiton Group for which traffic needs to be generated :

                1.Amazon-Apps                  2.Google-Apps            3.Social-Media-Apps            4.Office365-Apps        5.Exit
```

- Choose any option from the list (1-4) as necessary. This opens respective web-applications in a new browser tab.

For example, selecting "**4**" - Office365-Apps, will open all the listed web-apps (SharePoint, PowerPoint, Outlook, MS-Excel, MS-Teams) each in a new Chrome tab as shown below

```
Please choose the Predefined Applicaiton Group for which traffic needs to be generated :

                1.Amazon-Apps                  2.Google-Apps            3.Social-Media-Apps            4.Office365-Apps        5.Exit
4
Opening O365-Apps  "Sharepoint"  "Powerepoint"  "Outlook"  "MS-Excel" "MS-Teams"
```



- Similarly under "3.Application-Filter", we have the below predefined options:

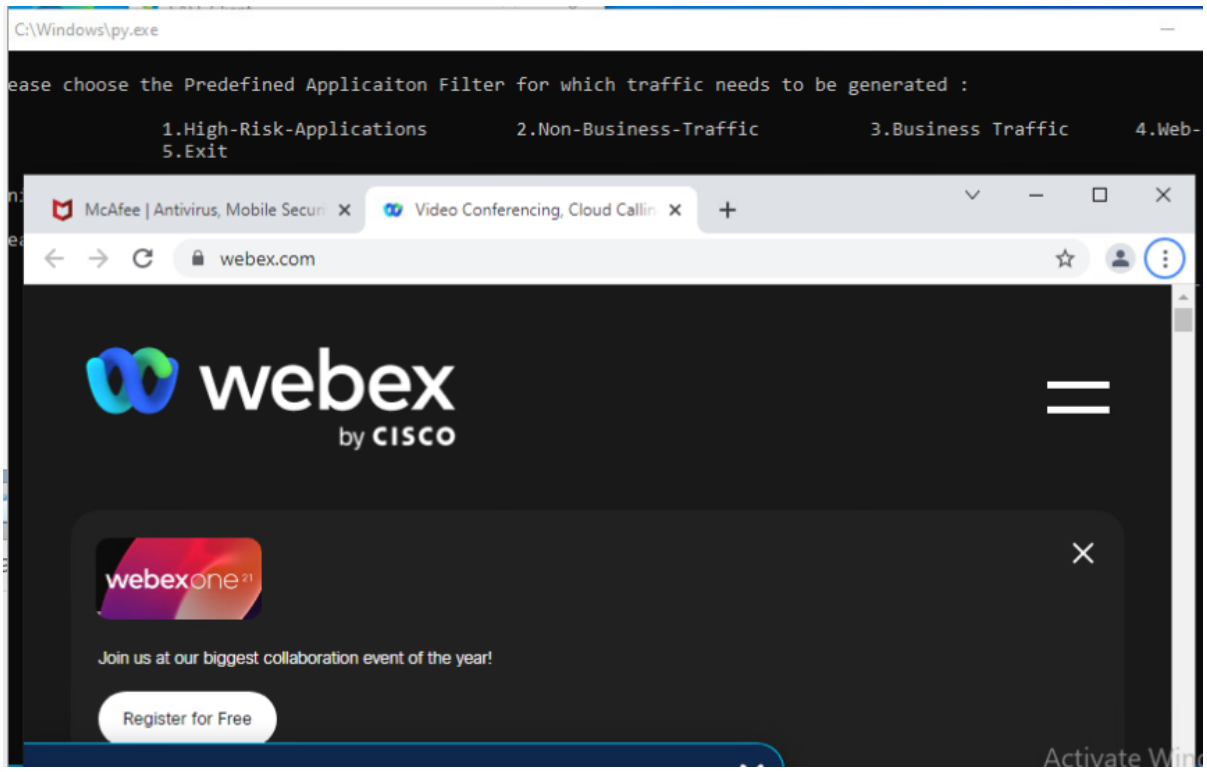
1. High-Risk-Applications
2. Non-Business-Traffic
3. Business Traffic
4. Web-Browsing

```
                1. Applications                2. Application Groups    3. Applicaiton Filter          4. Exit
3

Please choose the Predefined Applicaiton Filter for which traffic needs to be generated :
                1.High-Risk-Applications        2.Non-Business-Traffic          3.Business Traffic      4.Web-Browsing                5.Exit
```

- Choose any option from the list (1-4) as necessary. This opens respective web-applications in a new browser tab.

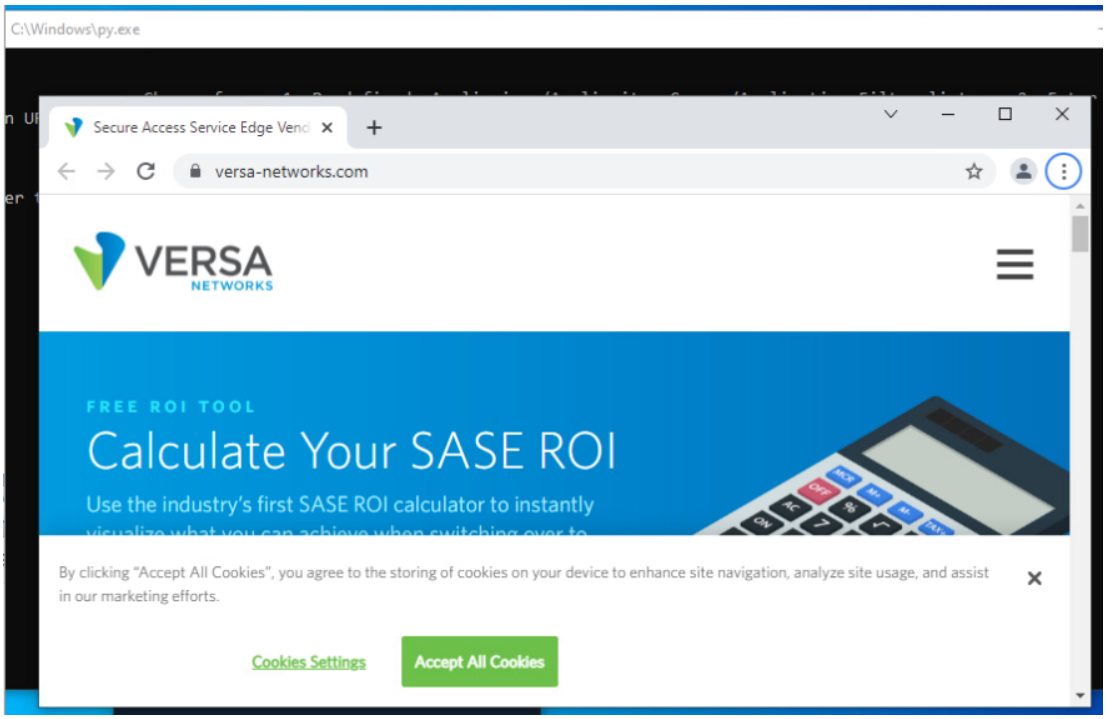
For example, selecting "3.Business Traffic" will open the below displayed applications(MacAfee, WebEx)in a new Chrome tab as shown below

```
                1.High-Risk-Applications        2.Non-Business-Traffic          3.Business Traffic      4.Web-Browsing                5.Exit
3
Opening Predefined Business Applications  "mcafee" "webex"
```



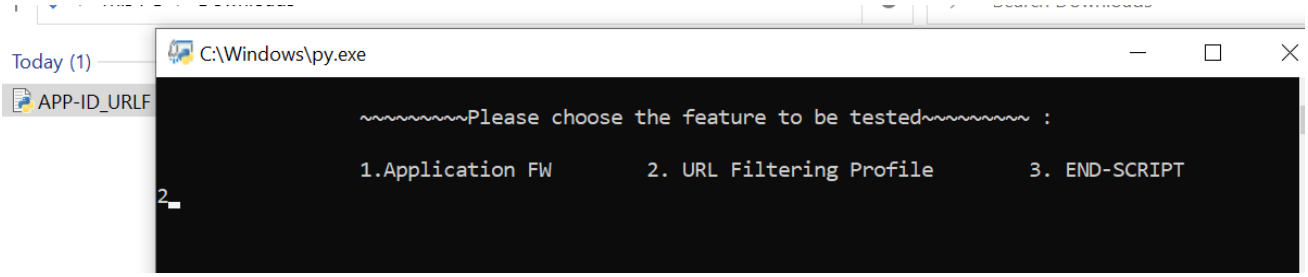
By selecting "2.Enter Web Application URL manually" from the main menu, script will prompt you to enter the URL Manually.

```
                Choose from : 1. Predefined: Applicaions/Applicaiton-Groups/Application-Filter list    2. Enter Web Application URL manually   3.E
2
                1. Enter the Application URL          2.Exit
1
Enter the URL :          https://www.versa-networks.com
```



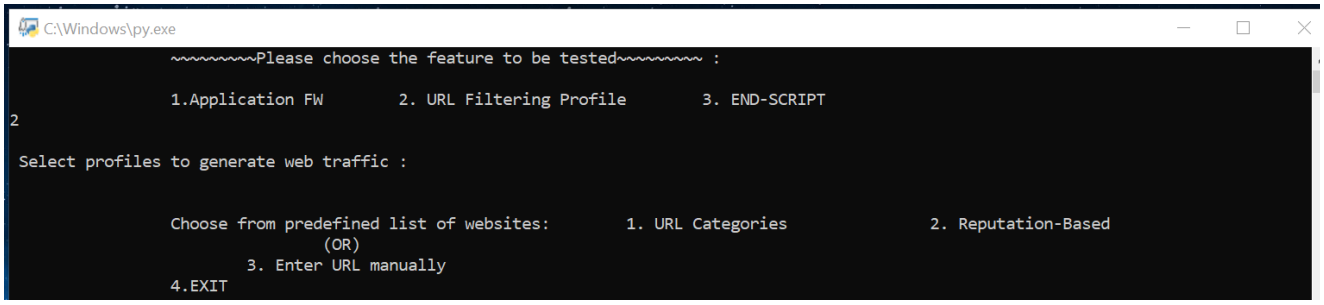
URL Filtering:

- Execute the script which opens up the python3 terminal as shown. For URL Filtering traffic profiles, type "**2** " in the prompt as below



- On pressing ENTER, it displays options for traffic-gen profiles. Either choose from predefined list (1-2) or type web-URL manually (3)

1. URL Categories
2. Reputation-Based, (OR)
3. Manual input



- Selecting "**1**" - URL Categories profile, prompts you to choose websites from one of the below categories:

```
Select category to open relevant webpages :

                1. Shopping            2. Social-Network       3. job_search   4. gambling     5. adult_and_pornography
                6. Malware sites       7. weapons              8. dating       9. auctions     10. proxy_avoid_and_anonymizers
                11. Goto Main menu
```

- Choose any URL-category from the list (1-10) as necessary. This opens the respective websites in a new browser tab.

  For example, on selecting "**1**" - Shopping, lets us choose any of the listed sites: 1. Amazon, 2. Flipkart, 3. Myntra

```
Select category to open relevant webpages :

                1. Shopping            2. Social-Network       3. job_search   4. gambling     5. adult_and_pornography
                6. Malware sites       7. weapons              8. dating       9. auctions     10. proxy_avoid_and_anonymi
                11. Goto Main menu
1
                1. Amazon              2. Flipkart             3. Myntra               4. Goto another category
```

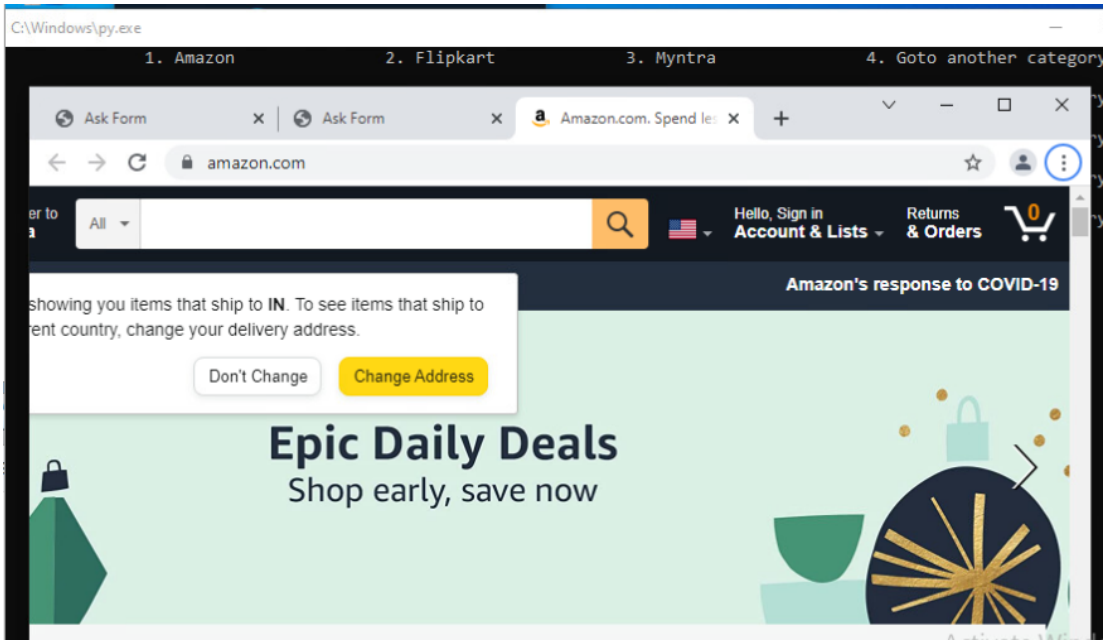
NOTE: In this demo set-up, URL firewall action for '*Shopping*' category is set to captive-action "*ASK*", whereas as an exception, *Amazon.com* website is configured to be *whitelisted* in URLF profile.

To verify this,

- Selecting "**1**" - Amazon in the terminal prompt,

```
Select category to open relevant webpages :

                1. Shopping            2. Social-Network       3. job_search   4. gambling     5. adult_and_pornography
                6. Malware sites       7. weapons              8. dating       9. auctions     10. proxy_avoid_and_anonymi
                11. Goto Main menu
1
                1. Amazon              2. Flipkart             3. Myntra               4. Goto another category
1
```

- As shown below, Amazon website opens in a new Chrome tab thus verifying URL whitelist action.



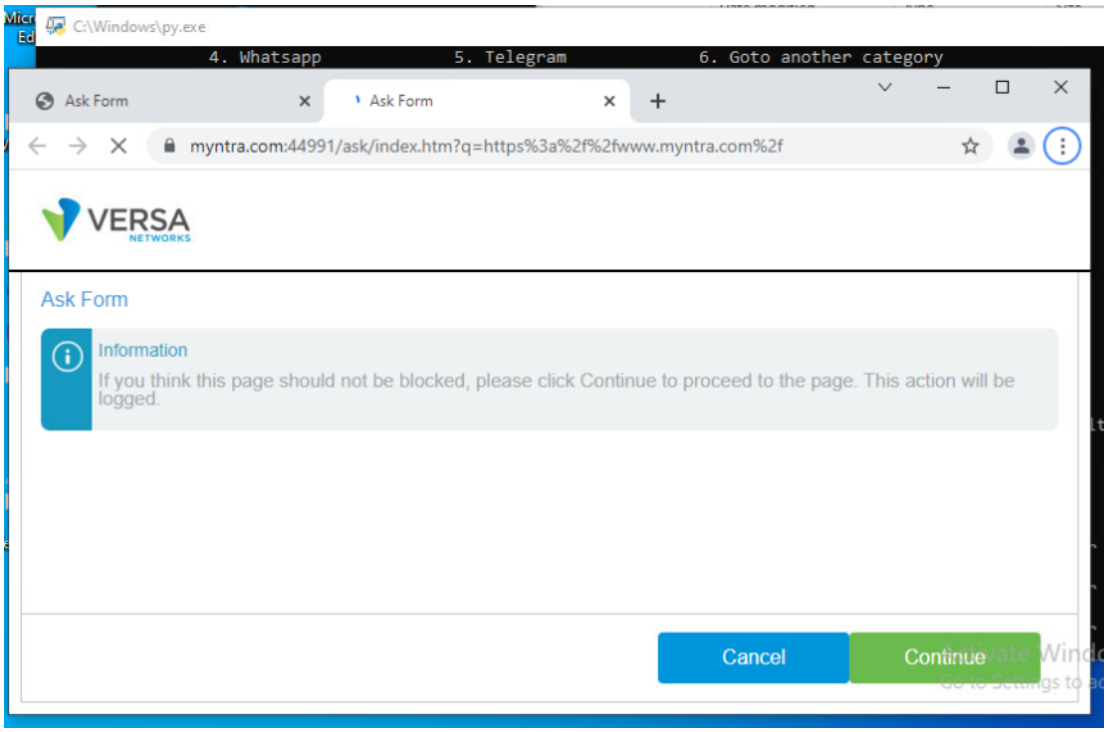
- Similarly to verify Caption-action "ASK" for other Shopping websites, choose "**3**" - Myntra in the prompt as shown,

```
Select category to open relevant webpages :

                1. Shopping            2. Social-Network       3. job_search   4. gambling     5. adult_and_pornography
                6. Malware sites       7. weapons              8. dating       9. auctions     10. proxy_avoid_and_anonymiz
                11. Goto Main menu
1
                1. Amazon              2. Flipkart             3. Myntra               4. Goto another category
3
```

- Selecting "**3**" will open Myntra webpage in a new Chrome tab.

  NOTE: Since captive-action is set to "*ASK*", the URL request redirects to captive-portal page with Ask Form as shown below.

  Clicking on 'Continue' (in captive-portal page) opens up Myntra website.

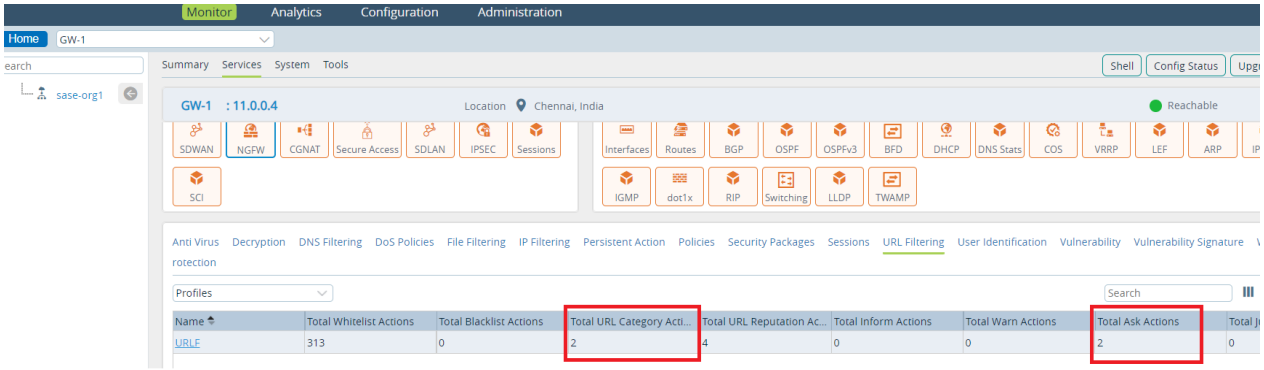


URLF profile rule hit-counts on VOS gateway can be verified using the command:

"show orgs org-services <Org-Name> security profiles url-filtering user-defined statistics <URLF-Profile-Name>"

```
security profiles url-filtering user-defined statistics URLF
 total_hits                          778
 total_url_string_matches            0
 total_url_regex_matches             299
 total_default_actions               473
 total_whitelist_actions             299
 total_whitelist_referrer_actions    0
 total_blacklist_actions             0
 total_blacklist_referrer_actions    0
 total_url_category_actions          2
 total_url_u_category_actions        0
 total_url_p_category_actions        2
 total_url_reputation_actions        4
 total_url_u_reputation_actions      0
 total_url_p_reputation_actions      4
 total_cloud_lookups                 18
 total_inform_actions                0
 total_warn_actions                  0
 total_ask_actions                   2
 total_justify_actions               0
 total_override_actions              0
```

The same can be viewed from Director GUI, go to the appliance & navigate to **Monitor -> Services -> NGFW -> URL Filtering**



- Similarly, any of the other URL-categories can be chosen to automatically open corresponding websites in a new Chrome tab & the Firewall rule actions on VOS device can be verified accordingly.

```
Select category to open relevant webpages :

                1. Shopping          2. Social-Network      3. job_search   4. gambling     5. adult_and_pornography
                6. Malware sites     7. weapons             8. dating       9. auctions     10. proxy_avoid_and_anonymizers
                11. Goto Main menu
```

NOTE: Category options 4-10 (as seen from above screenshot) currently have one website each & complaint in Versa SPACK 1893

Please try all profiles & if any suggestions please leave a comment

- Next under URLF traffic-gen profile we have **Reputation-based**

Reputation-Based traffic-gen profile has the following options:

1. Trustworthy
2. Moderate_risk
3. Suspicious
4. High-risk
5. Manual input

NOTE: Currently each risk-level has one website each & is complaint with Versa SPACK 1893

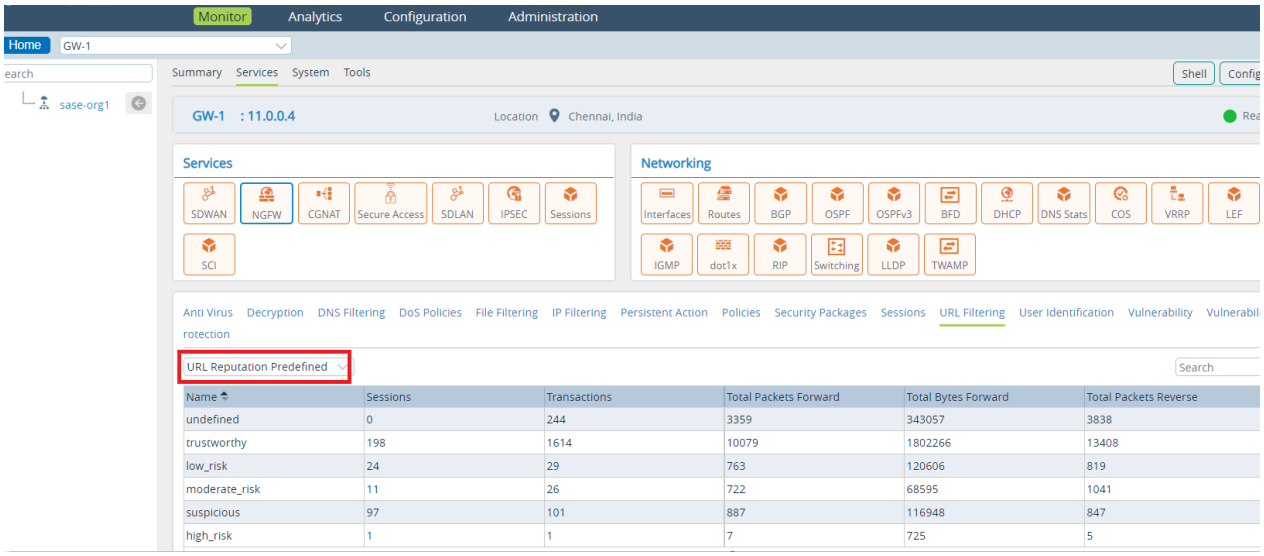
- To use this, choose "**2**" (Reputation-Based) from the sub-menu within 'URL Filtering Profile'

```
C:\Windows\py.exe
                ~~~~~~~~~Please choose the feature to be tested~~~~~~~~~ :

                1.Application FW        2. URL Filtering Profile        3. END-SCRIPT
2

 Select profiles to generate web traffic :


                Choose from predefined list of websites:        1. URL Categories              2. Reputation-Based
                                (OR)
                        3. Enter URL manually
                4.EXIT
2

 Please select Reputaion type to open relevant websites :


                1. Trustworthy          2. moderate_risk
                3. Suspicious           4. High-risk            5. Enter URL manually
                6. Goto Main menu
```

- From the displayed options (1-5), choose as desired to automatically open a corresponding website in a new Chrome tab

- Firewall rule actions corresponding to the websites accessed can be verified using the statistics command from VOS cli (or) from Director GUI under **Monitor -> Services -> NGFW -> URL Filtering** of the VOS appliance

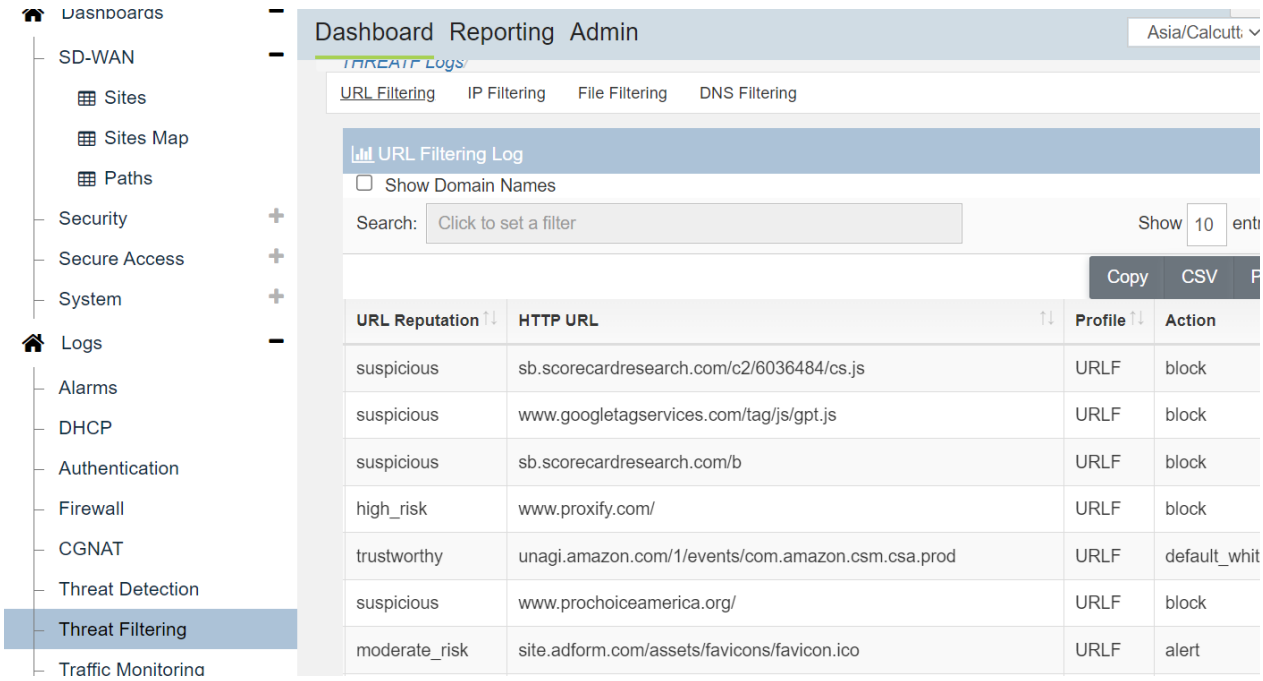
To view traffic statistics for predefined-reputations in VOS, select '*URL Reputation Predefined*' from the drop-down list as highlighted below:



NOTE: As per this set-up, on SASE GW, the URLF actions configured for each reputation are as follows: Trustworthy - "*ALLOW*", Moderate_risk - "*ALERT*", Suspicious - "*BLOCK*", High_risk - "*BLOCK*"

- The URLF profile statistics can also be verified from Analytics (provided LEF is configured)

  Below screenshot shows the URLF actions applied to the websites based on their reputation-level



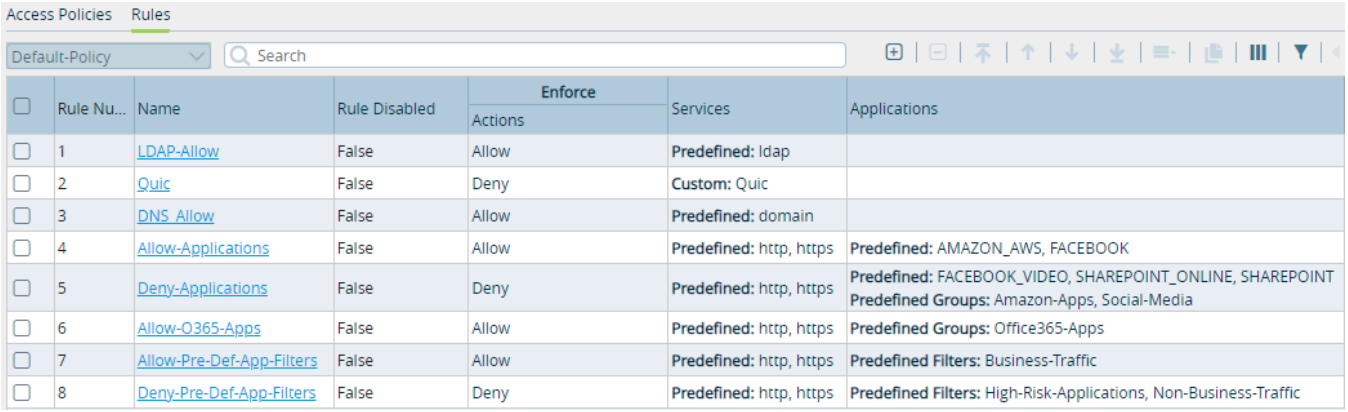
Configuration Template:

You can use this template as reference or import this predefined service template into your test environment & modify as necessary

Service templates(Type NextGen Firewall) -

- Application-Filtering.cfg

  The above Application-Filtering Service Template has the following Firewall policies with Application/App-Group/App-Filter Configured:



Rule Nu...	Name	Rule Disabled	Enforce Actions	Services	Applications
1	LDAP-Allow	False	Allow	**Predefined:** ldap	
2	Quic	False	Deny	**Custom:** Quic	
3	DNS_Allow	False	Allow	**Predefined:** domain	
4	Allow-Applications	False	Allow	**Predefined:** http, https	**Predefined:** AMAZON_AWS, FACEBOOK
5	Deny-Applications	False	Deny	**Predefined:** http, https	**Predefined:** FACEBOOK_VIDEO, SHAREPOINT_ONLINE, SHAREPOINT **Predefined Groups:** Amazon-Apps, Social-Media
6	Allow-O365-Apps	False	Allow	**Predefined:** http, https	**Predefined Groups:** Office365-Apps
7	Allow-Pre-Def-App-Filters	False	Allow	**Predefined:** http, https	**Predefined Filters:** Business-Traffic
8	Deny-Pre-Def-App-Filters	False	Deny	**Predefined:** http, https	**Predefined Filters:** High-Risk-Applications, Non-Business-Traffic

- URLF.cfg

  The above URLF Service template has the following URLF Profile configured:

Name	Deny List	Deny List Action	Allow List	Category Based Action		Reputation Based Action	
				Name	Actions	Name	Actions
URLF	**Pattern:** .*monster.*	**Predefined:** block	**Pattern:** .*amazon.*	Block_Sites Job_Search Shopping SocialNetwork	**Predefined:** block **Predefined:** allow **Predefined:** ask **Predefined:** justify	High_Risk Moderate_Risk Suspicious	**Predefined:** block **Predefined:** alert **Predefined:** block

NOTE-1: To reuse the service templates, replace the org name before importing to Director as below:

```
devices {
    template Application-Filtering {
        config {
            /* Tags: replace */
            orgs {
                org sase-org1 {
                    appliance-owner;      Replace with your ORG-NAME
                    services        [ nextgen-firewall ];
                }
                org-services sase-org1 {
                    security {
                        access-policies {
                            access-policy-group Default-Policy {
                                rules {
                                    access-policy LDAP-Allow {
                                        rule-disable false;
                                        match {
                                            source {
                                                user {
                                                    local-database {
                                                        status disabled;
                                                    }
                                                    external-database {
                                                        status disabled;
                                                    }
                                                    user-type any;
                                                }
                                            }
                                            services {
                                                predefined-services-list [ ldap ];
                                            }
                                        }
                                        set {
                                            action                allow;
                                            tcp-session-keepalive disabled;
                                            lef {
                                                event never;
                                                options {
                                                    send-pcap-data {
                                                        enable false;
                                                    }
                                                }
                                            }
```

NOTE-2: Above service template does not include SSL Decryption Rule & Captive-portal configuration. Please make sure the same is configured in your test environment for captive portal actions (Decryption rule with action 'decrypt' or 'no-decrypt' will work).